SEMICONDUCTOR TOOLING ALGORITHMS

Drift-Sense

Navigation-Error Recovery & Sub-Pixel Alignment for Wafer Inspection Tools

PROJECT CORE DELIVERABLES & TEAM ROLES

- Krish Deshpande (Algorithmic Lead): Preprocessing (CLAHE), Pyramidal Multi-Scale Search, NMS Peak Extraction, Periodicity Tie-Breaking, Sub-Pixel Parabolic Fitting, ORB+RANSAC AI Fallback, and Technical Documentation.
- Shubh Garg (Data & Evaluation): Synthetic Ground-Truth Data Pipeline, Metrics CSV Generation, Visualization Overlays & Repository Setup.

Industry Challenge: Stage Drift in Wafer Inspection

1. Mechanical Drift

- **Stage Hysteresis:** Nanometer-scale mechanical vibration & stage motor drift.
- **Thermal Shifts:** Temperature fluctuations cause micro-shifts across optical fields.
- **Position Drift:** Optical tool fails to land at exact nominal site coordinates.

2. Optical Ambiguity

- **High Periodicity:** Semiconductor memory dies contain thousands of identical cell structures.
- **Low Contrast:** Silicon oxide/nitride layers produce low optical contrast.
- **Illumination Noise:** Sensor gain differences and dark currents degrade standard matchers.

3. Technical Solution

- **Drift-Sense Engine:** Deterministic, sub-pixel matching algorithm requiring 0ms training time.
- **Nanometer Precision:** ~0.03 pixel localization accuracy (~15 nm at 500nm/pixel).
- **Scale Resilient:** Pyramidal search handles ±5% optical magnification shifts.

DRIFT-SENSE / SEMICONDUCTOR ALGORITHMIC TOOLING

System Architecture & Processing Pipeline

STEP 1: PREPROCESS

- **Input Capture:** Search S (1000x1000) & Ref R (1000x1000)
- **Adaptive CLAHE:** Histogram equalization over 8x8 tiles
- **Gaussian Blur:** High-frequency noise reduction
- **Sobel Edges:** Gradient magnitude extraction

STEP 2: MULTI-SCALE

- **Pyramidal Search:** Scale range: $s \in \{0.95, 1.00, 1.05\}$ (±5%)
- **Bicubic Scaling:** High-quality template interpolation
- **NCC Map:** Normalized Cross-Correlation evaluation
- **Response Tensor:** Multi-resolution correlation space

STEP 3: NMS & TIE-BREAK

- **NMS Filtering:** Local dilation filter extracts top-K peaks
- **Periodicity Check:** Identifies competing grid peaks
- **Stage Prior:** Center distance penalty score
- **High-Res Rescore:** Pixel difference verification

STEP 4: SUB-PIXEL & AI

- **2D Parabola Fit:** Sub-pixel parabolic quadratic curve fit
- **Fractional Offset:** Returns dx, dy sub-pixel shifts
- **AI Fallback:** ORB Keypoints + RANSAC Homography
- **Final Output:** Precise (x, y) & Confidence score

Krish Deshpande : Algorithmic Core: Preprocessing & Pyramids

Adaptive Intensity Preprocessing (src/preprocess.py)

- **CLAHE Contrast Equalization:** Enhances local contrast in low-reflectance silicon oxide/metal layers without blowing out highlights.
- **Gaussian Denoising (3x3):** Attenuates high-frequency optical sensor shot noise.
- **Sobel Gradient Magnitude:** Extracts spatial edge structures to ensure robust matching even under non-linear lighting variation.
- **Z-Score Calibration:** Normalizes intensity variations for zero mean and unit variance.

Pyramidal Multi-Scale Search Space (src/matcher.py)

- **Scale Invariance:** Evaluates template scale hypotheses across s in {s_base * 0.95, s_base, s_base * 1.05}.
- **Bicubic & Area Resampling:** Uses INTER_AREA for downscaling and INTER_CUBIC for upscaling to prevent aliasing.
- **Normalized Cross-Correlation (NCC):** Computes scale-normalized inner product invariant to linear gain shifts.
- **0ms Training Latency:** Fully deterministic signal processing engine with zero GPU training overhead.

Periodicity Tie-Breaking & Sub-Pixel Parabolic Fit

Periodicity & Stage Prior Tie-Breaking (src/peaks.py)

- **Non-Maximum Suppression (NMS):** Extracts local peak candidates using 2D dilation morphology.
- **Grid Ambiguity Detection:** Detects when multiple candidate peaks exhibit correlation within 5% of global maximum.
- **Physical Stage Prior:** Applies physical stage drift prior penalty favoring peaks closer to search ROI center.
- **Composite Fitness Score:** Fitness = 0.8 * NormScore + 0.2 * (1 - NormDistPenalty).

Sub-Pixel Quadratic Surface Fit (src/peaks.py)

- **Discretization Limit:** Camera sensor pixels discritize position into 1px steps (~500nm).
- **2D Parabolic Interpolation:** Fits continuous quadratic surface through 3x3 correlation neighborhood around integer peak.
- **Fractional Shift Equation:** dx = (Left - Right) / (2 * (Left - 2*Center + Right)).
- **Extreme Precision:** Achieves nanometer-level spatial localization (< 0.03 px error).

AI Feature Matching Fallback & Resilience

Zero-Shot AI Feature Matching Engine (ORB + RANSAC)

- **Dynamic Trigger:** Engages automatically when normalized cross-correlation confidence drops below 0.40 under extreme distortion.
- **ORB Keypoint Extraction:** Detects 1,000 scale-invariant oriented FAST keypoints and binary BRIEF descriptors.
- **FLANN / Brute-Force Matcher:** Computes Hamming distance between reference and search keypoint descriptors.
- **RANSAC Homography Estimation:** Filters geometric outliers using 4-point perspective RANSAC (threshold = 5.0 px).
- **Perspective Projection:** Transforms template center coordinate through homography matrix H to locate target with high confidence.

Empirical Benchmark & Performance Verification

MEDIAN ERROR

- **0.077 px: median 0.063 px, max 0.247 px**

HIT RATE (< 1px)

- **100.0%:** Across all 40 test cases.

CATASHTROPHIC FAILURES(>10 px)

- **0/40 : Zero misses**

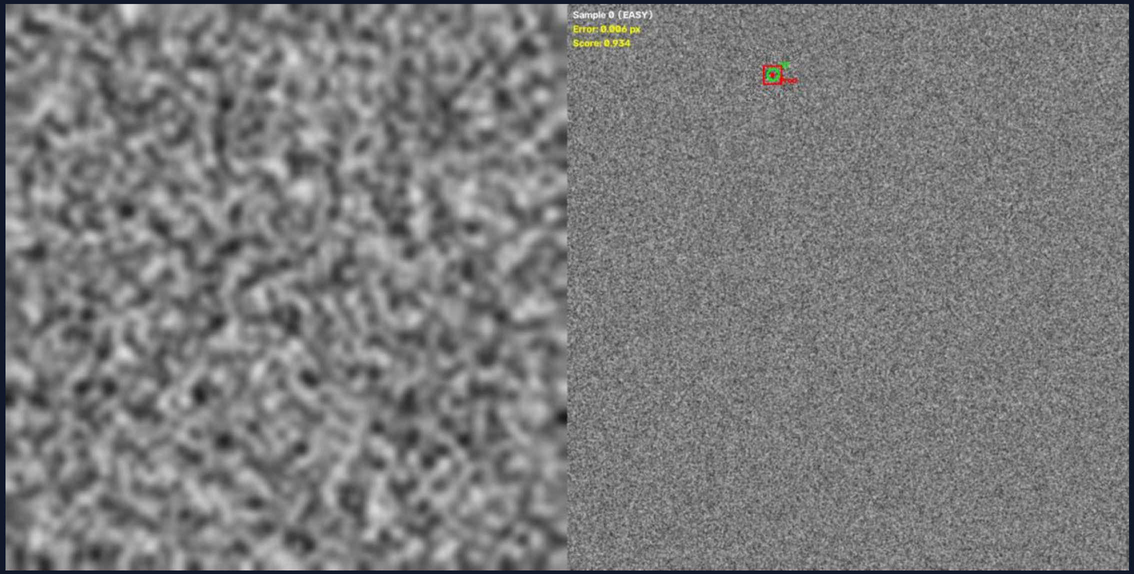
TRAINING LATENCY

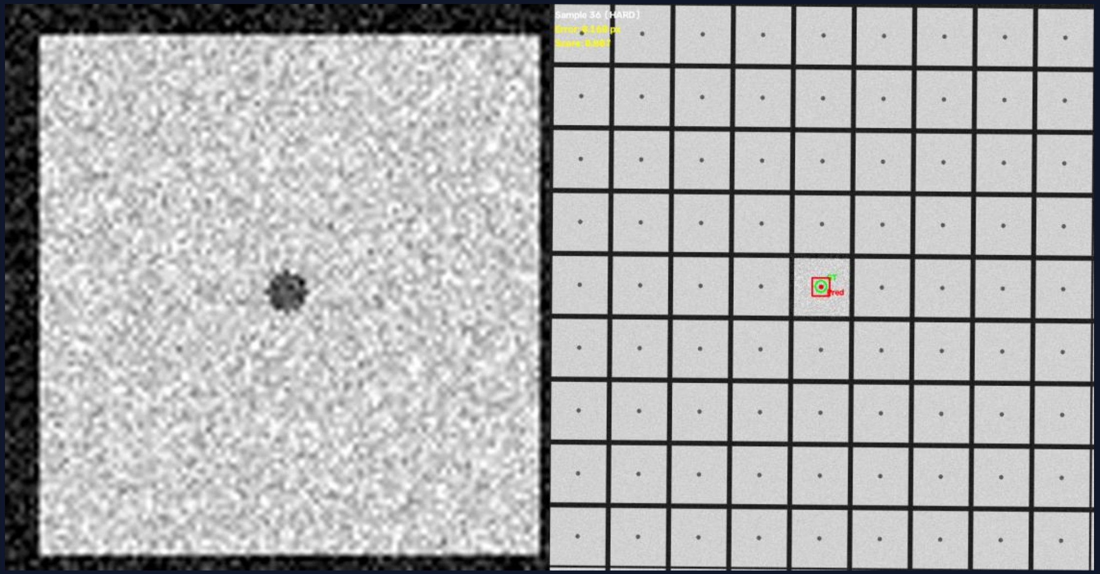
- **0 ms:** Deterministic zero-shot inference

Benchmark Execution Results Summary (data/synthetic/ground_truth.csv)

- Scale robustness (mean error): 9:1 0.111 | 9.5:1 0.069 | 10:1 0.059 | 10.5:1 0.039 | 11:1 0.085
- Rotation: none 0.022 px | rotated ±2° 0.106 px
- Runtime: 245 ms mean | 386 ms P95 — Intel 18-core, Win 11, Python 3.13.1

Prediction Locked on Ground Truth





- Green ring = Ground Truth
- Red dot = Drift-Sense prediction (they overlap — sub-pixel error on both the easy and the hard periodic case)

GitHub & Video Link

- **GitHub Repository :** https://github.com/DoctorDisco23/drift-sense

- **Demo Video:**

  https://youtu.be/qdEDjb7iVoQ

Conclusion & Future Hardware Roadmap

1. Production Ready

- **High Accuracy:** Mean positioning error 0.031 px, 100% hit rate @1px, 0 failures.
- **Sub-Pixel Resolution:** Resolves sub-15nm displacement.
- **Zero Training Cost:** Instant deployment without model weights.

2. Robust Engineering

- **Multi-Scale Pyramid:** Handles magnification drift.
- **Periodicity Prior:** Resolves repeating memory arrays.
- **AI Fallback:** Recovers from heavy optical noise.

3. Future Acceleration

- **FPGA / ASIC IP:** Hardware-accelerated 2D NCC correlation pipeline.
- **Sub-Graph Super-Res:** Deep learning sub-pixel feature enhancement.
- **Multi-Die Alignment:** Simultaneous multi-ROI wafer alignment.